ETSI STQ Conference 2025



Comparing Prediction Models for Perceived Listening Effort

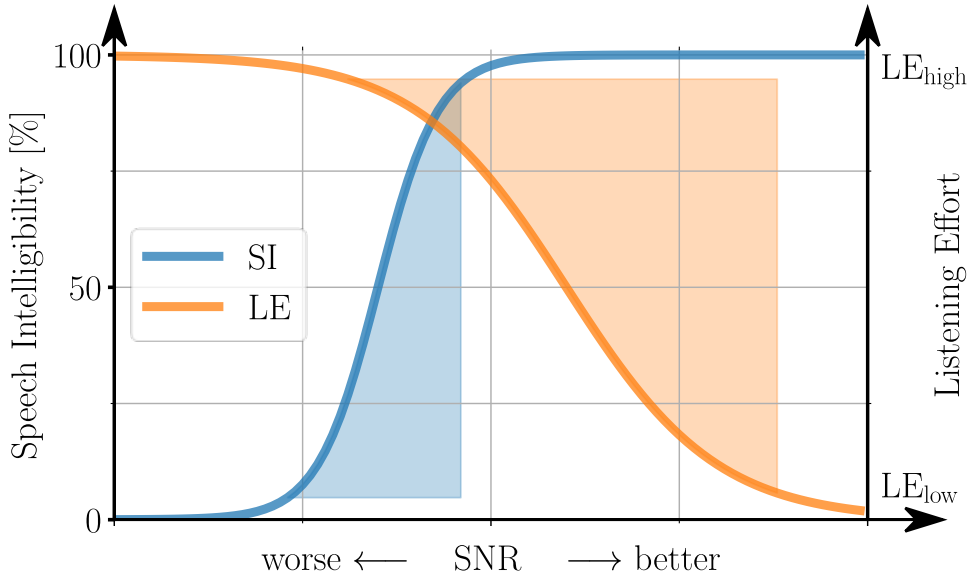
Presented by: Jan Reimes

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14/05/2025

Introduction

- Speech communication is getting more complicated …
- *Listening Effort* subject of various activities in research & industry
- Advantages over measuring *Intelligibility:*





- Demand for prediction models that are accurate, robust, versatile …

Prediction Models ETSI TS 103 558

- "Methods for obj



- Double-ended p
- Multiple metrics
- Formally validat
  → focus on qual

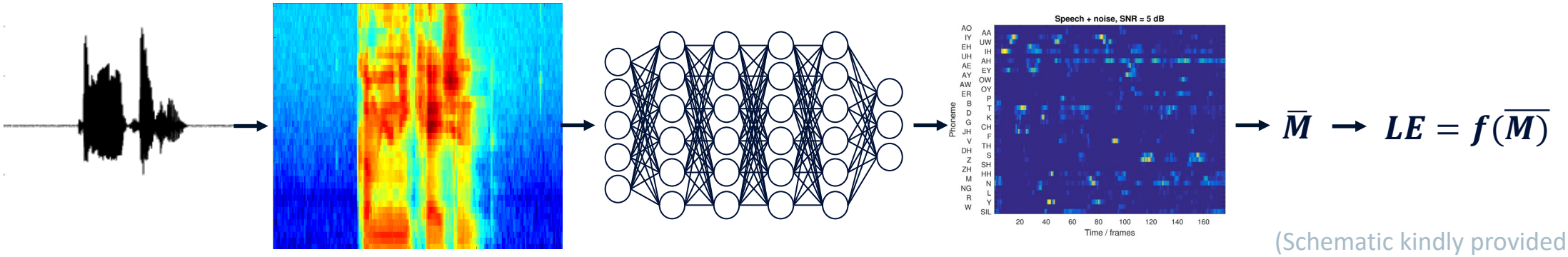
ETSI TS 103 558 V1.3.1 (2021-07)

TECHNICAL SPECIFICATION

Speech and multimedia Transmission Quality (STQ);
Methods for objective assessment of listening effort

Prediction Models – LEAP

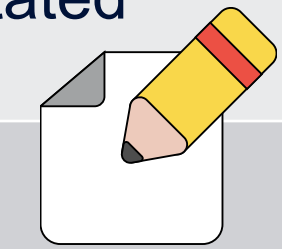
- Listening Effort Prediction from Acoustic Parameters
  ➔ [Rennies2022], [Huber2018], [Hermansky2013], et al …



(Schematic kindly provided by Bernd T. Meyer)

- Single-ended model – no reference signal required
- Utilizes speech recognition backend / intermediate output → "posteriorgrams"
- Degree of "smearing" quantified by "Mean Temporal Distance" ("M-Measure")
- Binaural assessment via better ear

Prediction Models - Comparison

	TS 103 558 / ABLE	LEAP
Applications	Scenarios with access to electro-acoustic signal insertion	Live/synthetic speech, non-intrusive scenarios
Speech signals	Phonetically balanced ("Harvard-like")	Any (?)
Test design	Categorial / opinion test ✓	Categorial / opinion test
Trained on	Subjective LE data 	Phoneme-annotated speech 
Training data	~ 10k samples	~ 8k hours

Prediction Models – Auditory scales

- Both models based on categorial tests ✓
- … but use different scales

ABLE: Mean Opinion Score (MOS_{LE})
Worst = 1 → Best = 5

Effort required to understand the meanings of sentences	*Score*
Complete relaxation possible; no effort required	5
Attention necessary; no appreciable effort required	4
Moderate effort required	3
Considerable effort required	2
No meaning understood with any feasible effort	1

LEAP:
Effort Scale Categorical Units (ESCU)
Worst = 14 → Best = 1

How much effort does it require for you to follow the speaker?
only noise
extreme effort
:
very much effort
:
considerable effort
:
moderate effort
:
little effort
:
very little effort
:
no effort

- → non-linear shifts in predictions due to scales (+ context effects)

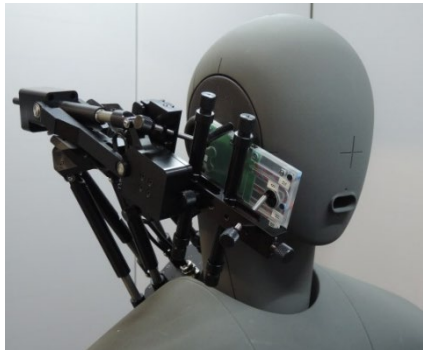
Listening Effort – Comparison – Data

Auditory validation databases available in ETSI TS 103 558 (Annex D):

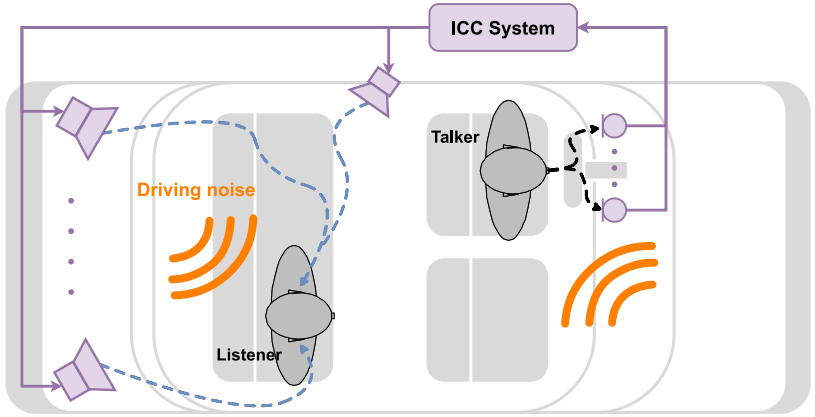
- Large-scale experiments of "speech in noise"
  → various & heterogeneous scenarios/devices
- Binaural recordings in different languages
- Auditory assessment of MOS_{LE}
- Data unseen to both models!



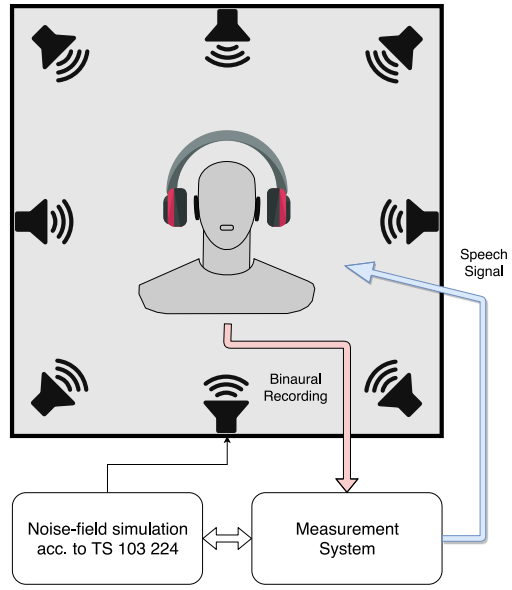
Handheld hands-free



Handset



In-car communcation



ANC-headsets

Listening Effort – Comparison – Data

Application	Lang.	# conditions	Σ samples
ANC-headsets [ANC]	ENG	48	384
In-car communication [ICC]	MAN	48	384
Handheld hands-free [HF]	MAN	24	192
Handset [HS]	GER	24	192

- 32-48 listeners (NH)
- 16 votes/sample
- 8 samples/condition
- ➔ 128 votes/condition

Metrics for prediction performance:

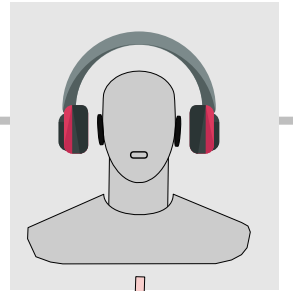
- rmse*
- max. abs* (worst-case/outlier)

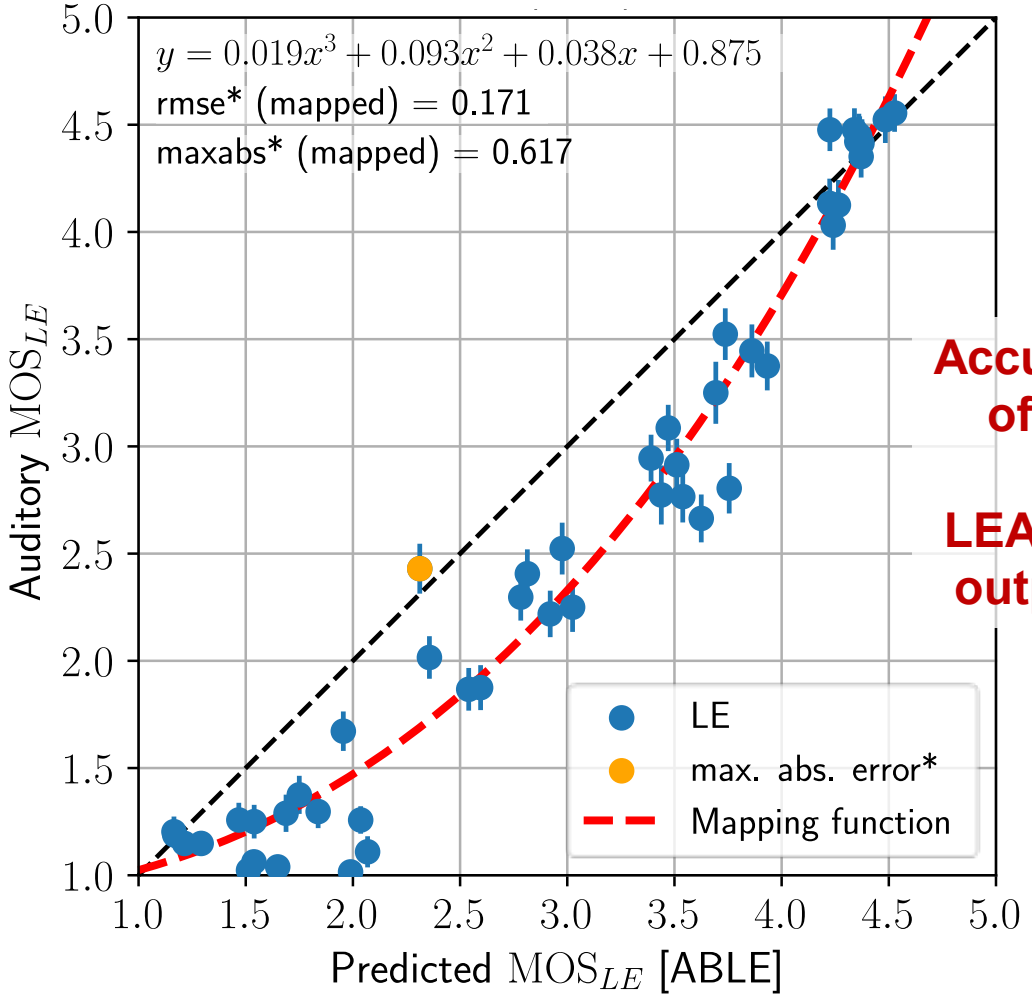
→ ITU-T Rec. P.1401

* = After …

- applying 3rd order mapping function
  → Context effects, different scales, …
- Compensating for uncertainty of subj. data:
  $d^* = \max(0, |x_{obj} - x_{subj}| - ci95)$

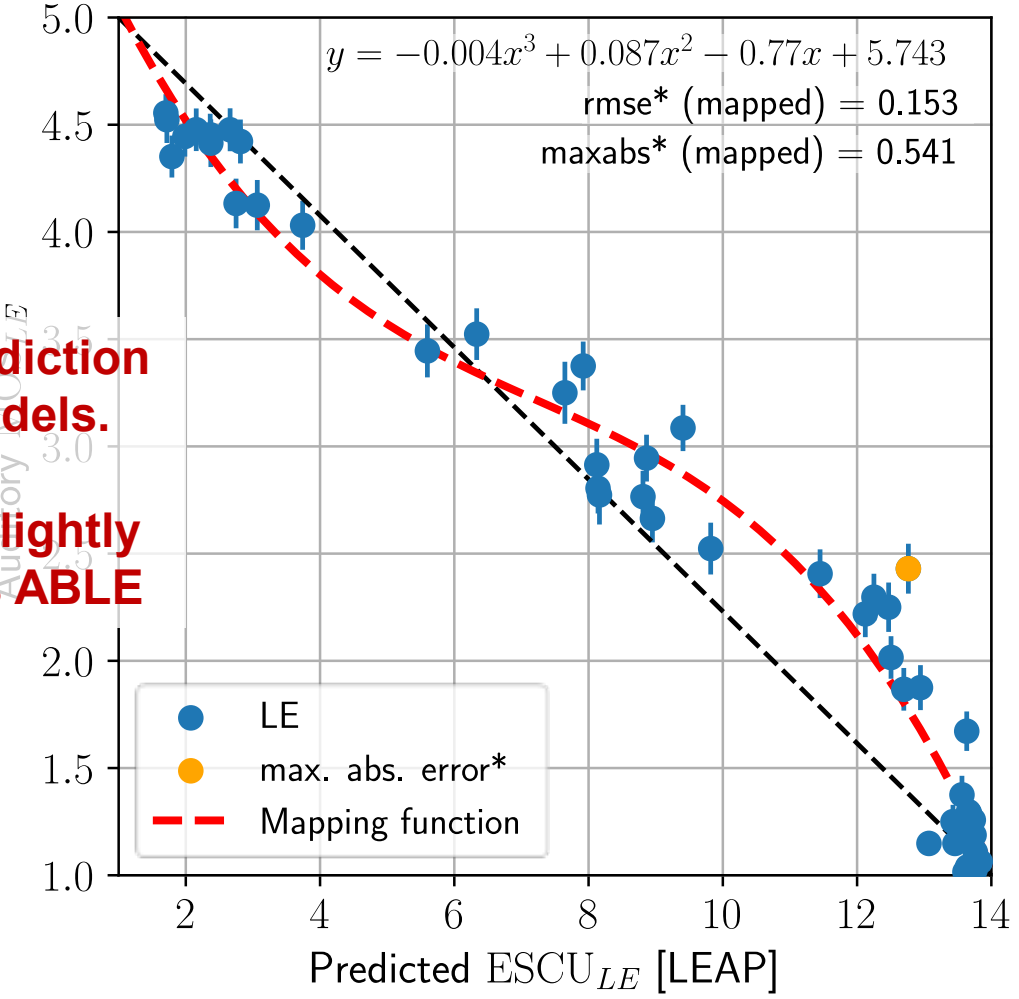
Listening Effort – Comparison – ANC-Headsets (English)





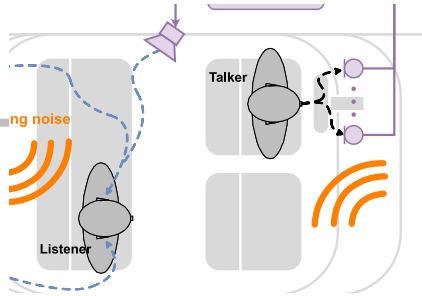
Accurate prediction of both models.

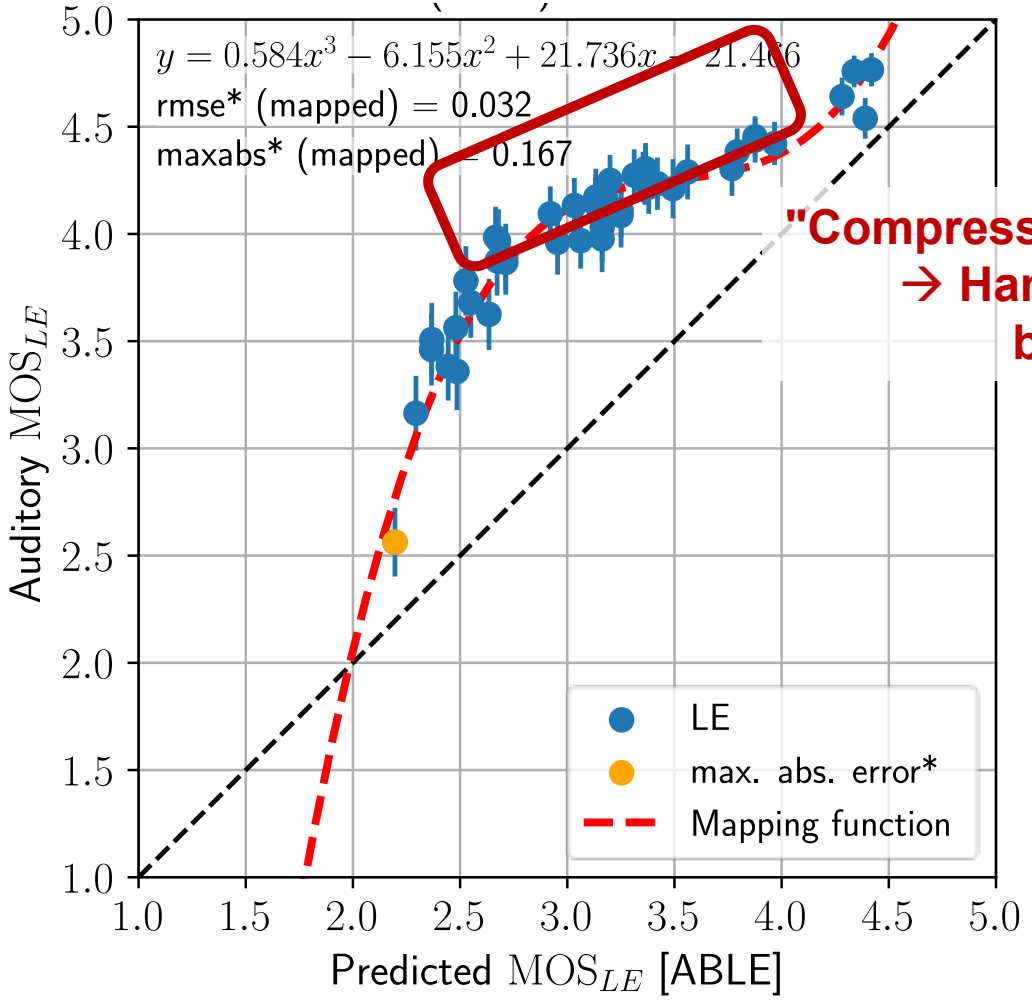
LEAP even slightly outperforms ABLE



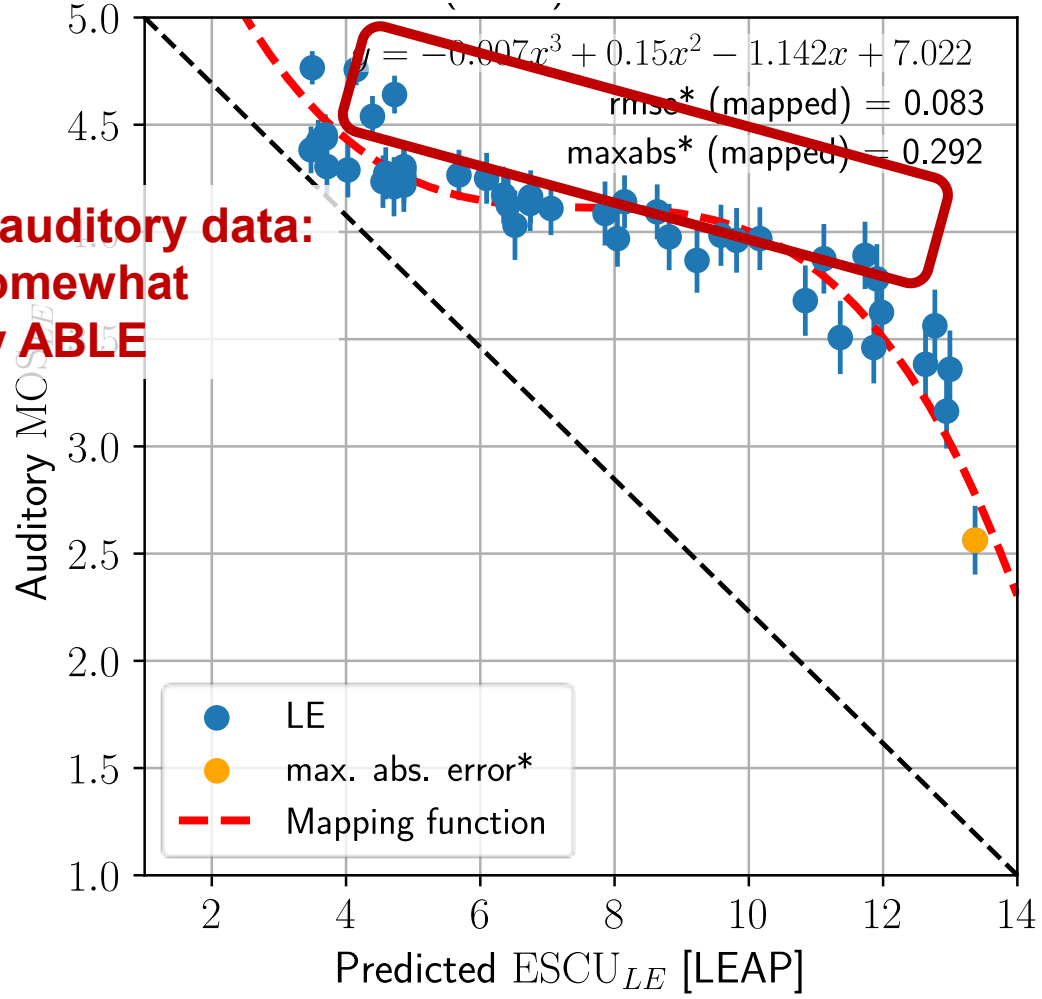
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Listening Effort – Comparison – ICC (Mandarin)



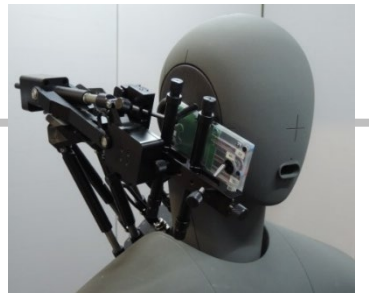


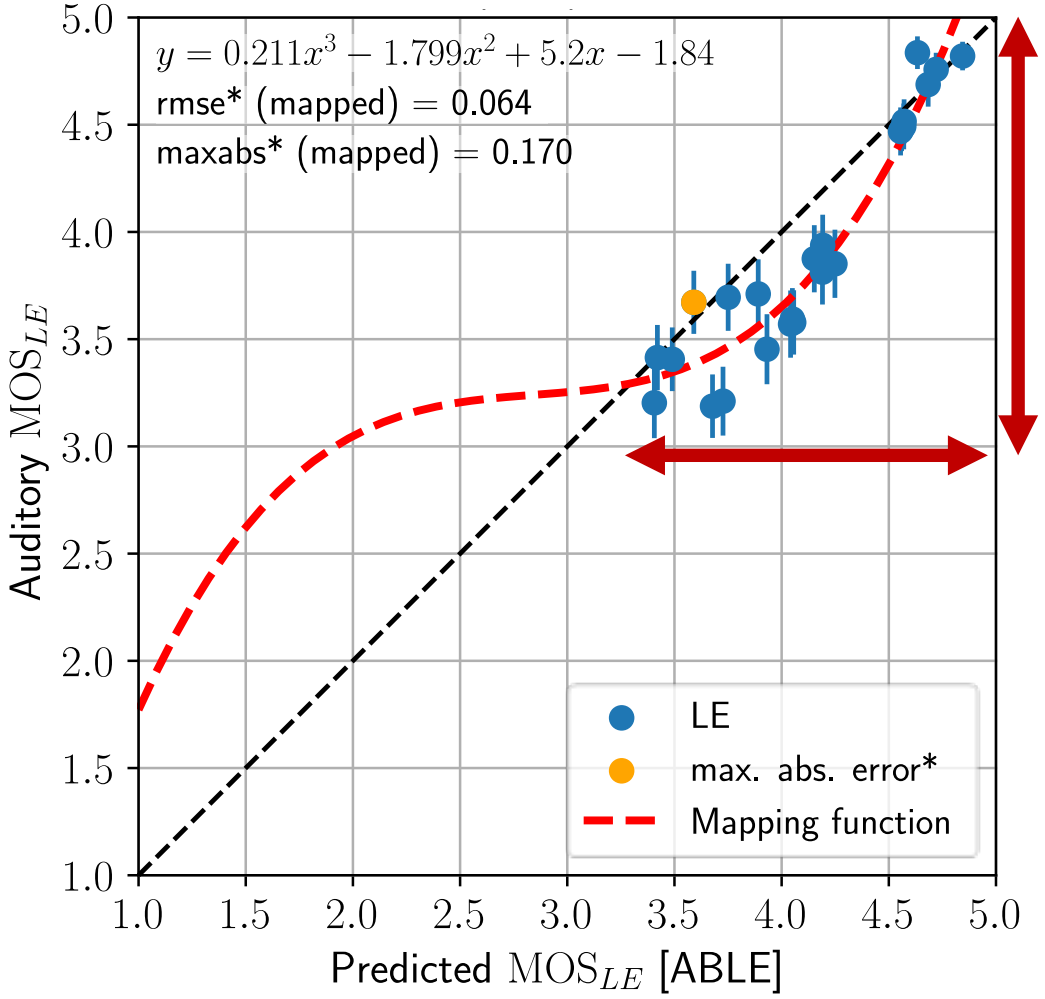
"Compression" in auditory data:
→ Handled somewhat better by ABLE

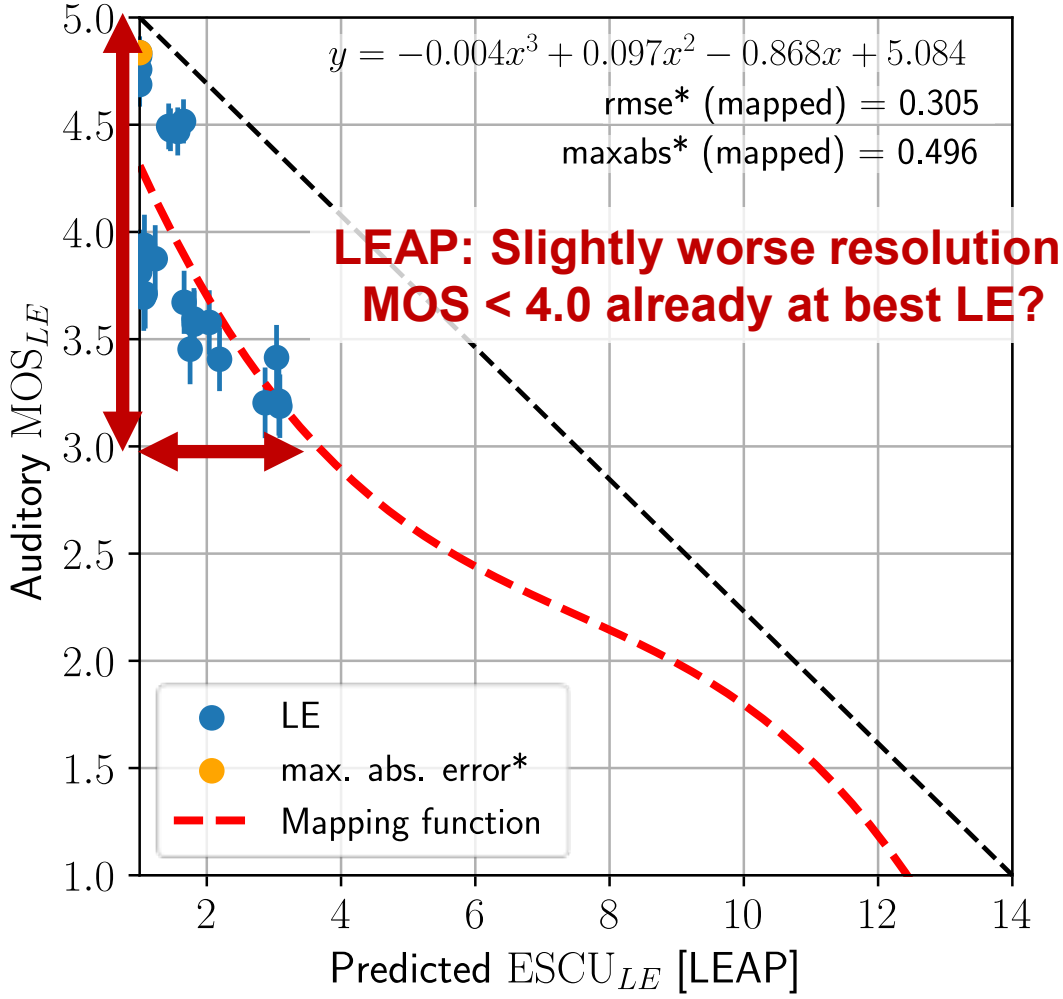


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Listening Effort – Comparison – HS (German)



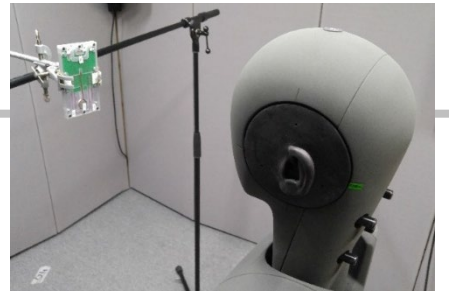


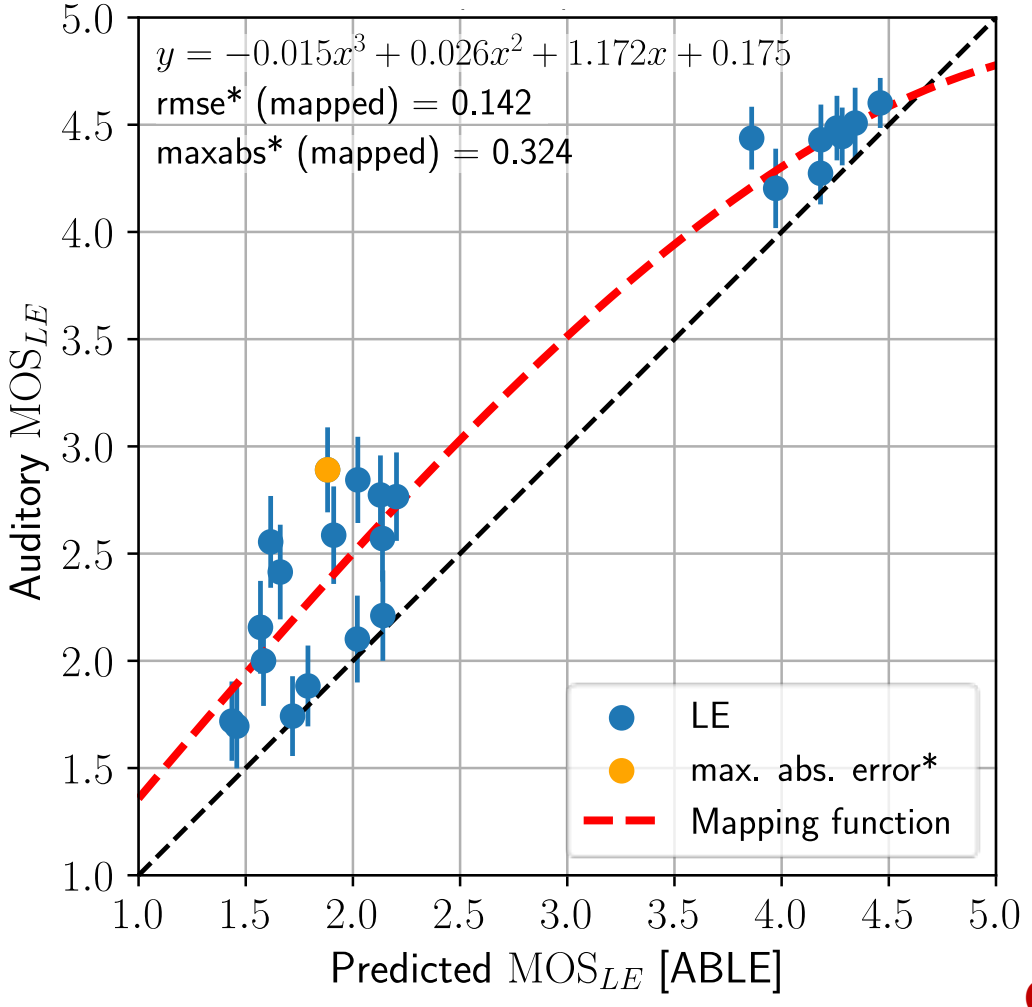


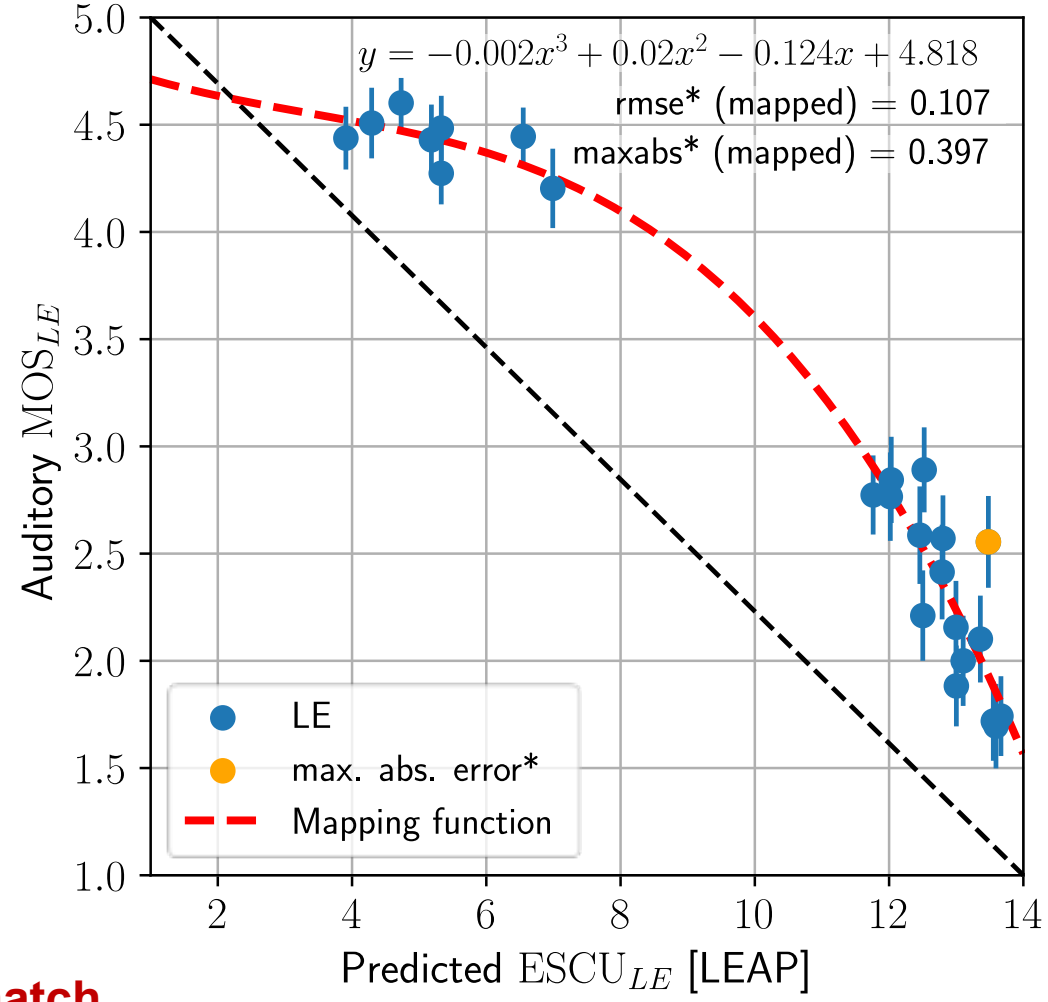
LEAP: Slightly worse resolution, MOS < 4.0 already at best LE?

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Listening Effort – Comparison – HHHF (Mandarin)







Good match

Summary & Conclusion

- Two prediction models for perceived listening effort investigated
  - Both based on categorial opinion tests …
  - … but with completely different approaches

- Prediction performance in various acoustic scenarios of …
  - ABLE: accurate for validated scenarios ("as officially reported")
  - LEAP: seems to generalize well for quite different use cases & different languages (phonemes trained on German speech only)

- Outlook / further investigations on …
  - Systematic differences (e.g., common versus different mispredictions)
  - More applications (e.g., for synthetic speech → data from [Huber2022])

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